Synergistic Motifs in Linear Gaussian Systems

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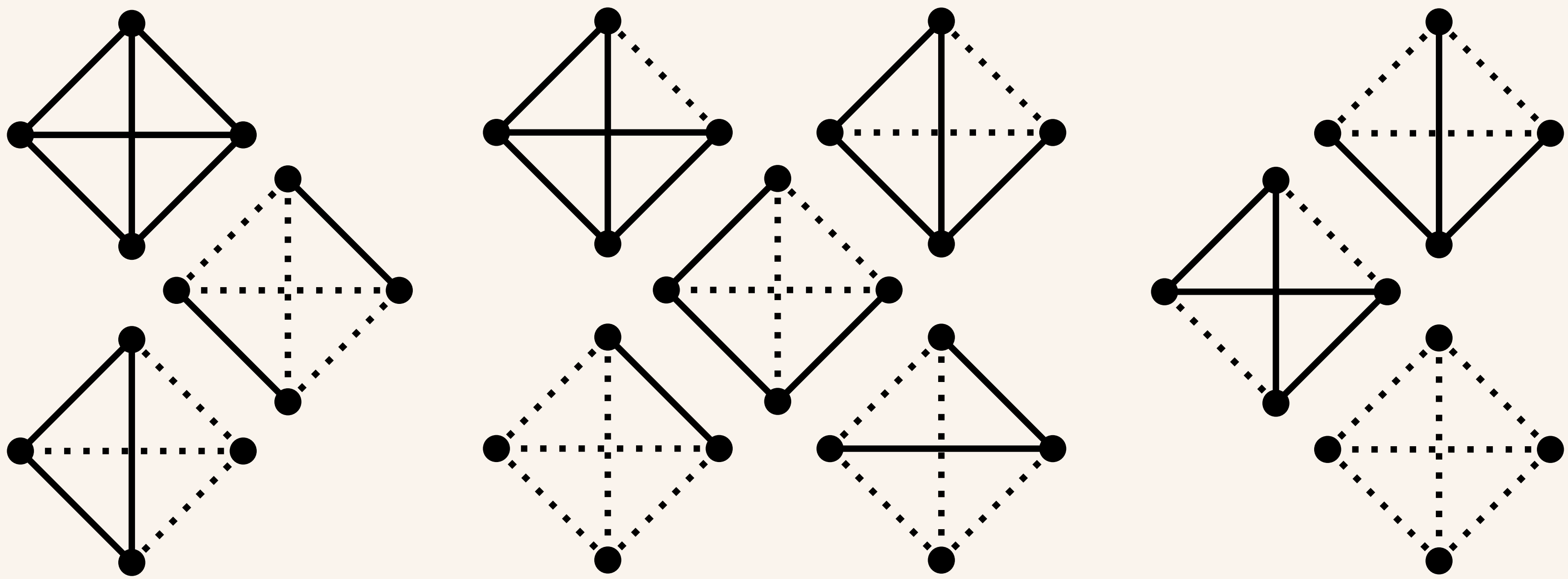


Question

What kind of mechanisms are needed for higher-order interdependencies to arise?

What about low-order mechanisms? (i.e., pairwise interactions)

1. What do synergy-dominated Gaussian Systems *look like*?
2. What are the simplest mechanisms that can lead to these correlational structures?

Outline

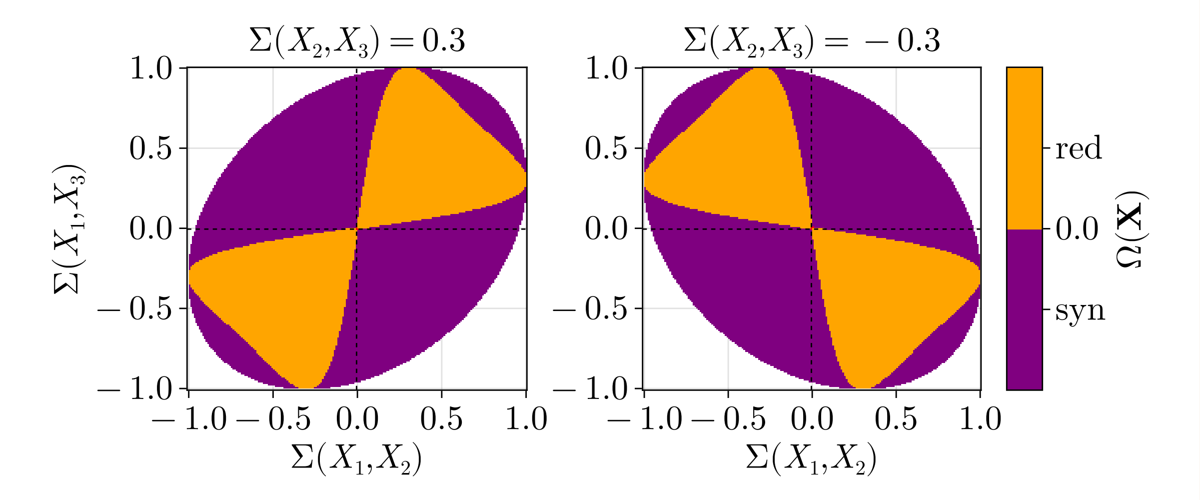
- *Partial Information Decomposition* (I trust someone has already introduced this)
- a little bit of *Structural Balance Theory*
- some *Gaussian Information Theory*
- Results!
- empirical validations, limitations etc

Intuition

From Barrett (2015):

$$\Sigma = \begin{pmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{pmatrix} \quad \rightarrow \quad \Omega(\boldsymbol{X}) = \frac{1}{2}\ln\left[\frac{1-(a^2+b^2+c^2)+2abc}{(1-a^2)(1-b^2)(1-c^2)}\right]$$

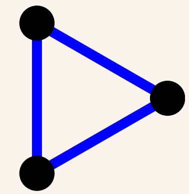
$\Rightarrow \Omega < 0$ **if** $2abc < (ab)^2 + (ac)^2 + (bc)^2 - (abc)^2$



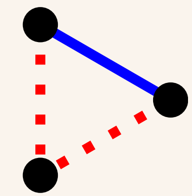
Barrett (2015) "Exploration of synergistic and redundant information sharing in static and dynamical Gaussian systems" Physical Review E

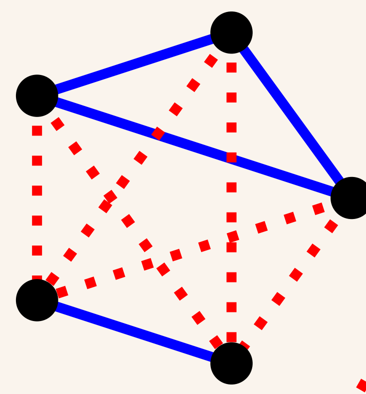
Structural Balance Theory

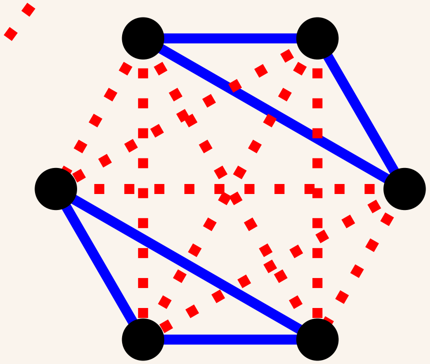
Intuition:



- the ***friend*** of my ***friend*** is my ***friend***
- the ***enemy*** of my ***friend*** is my ***enemy***







Note:

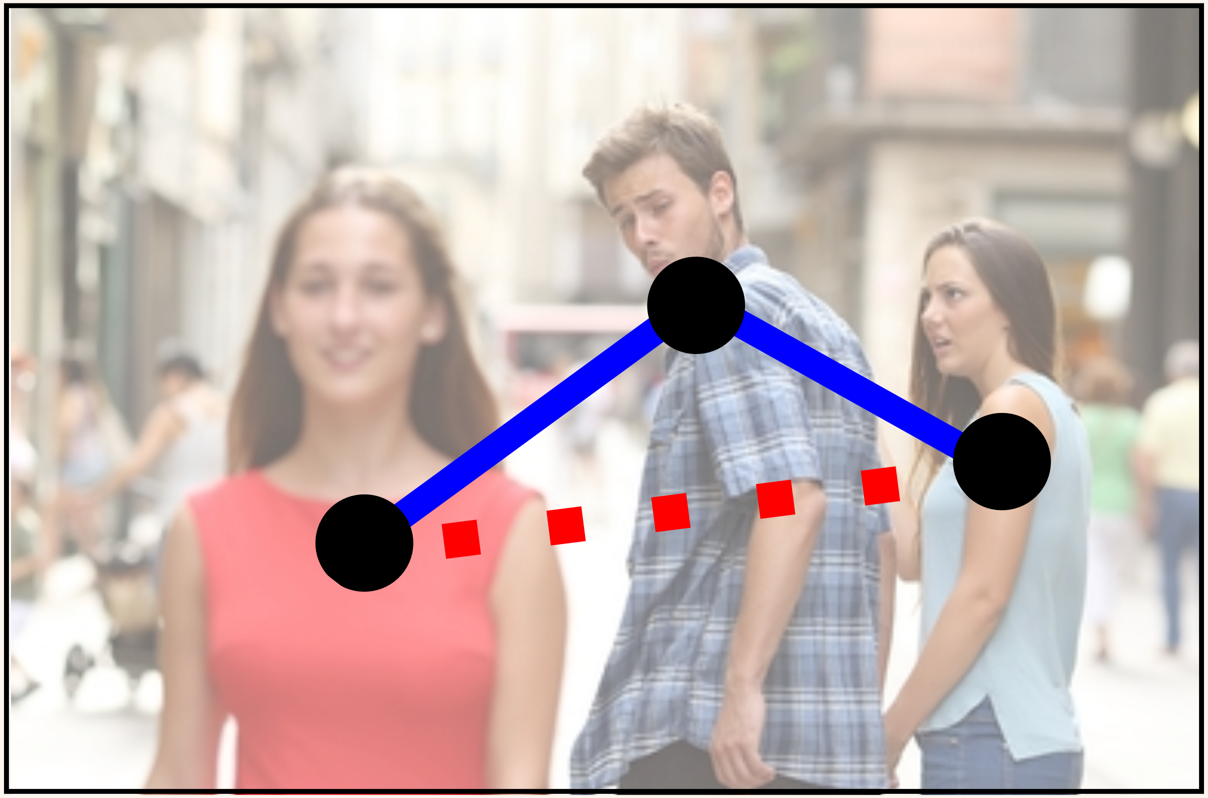
$(A^3)_{ii} > 0$ **or in our notation**

$\forall w \in \boldsymbol{w}^3, \quad d(w) > 0$

where $w = (i_1, i_2, \ldots, i_k, i_1) \in \boldsymbol{w}^k$

and $d(w) = \prod_{i=1}^{k} A_{w_i, w_{i+1}}$

Cartwright, D.; Harary, F. (1956). "Structural balance: a generalization of Heider's theory" Psych. Review.

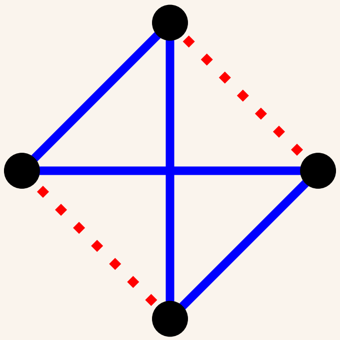


Structural Duality

Structure theorem for antibalance

a graph G is antibalanced if it can be partitioned into two non-overlapping sets of vertices, V_1 and V_2 (at least one non-empty), such that edges between vertices in the same set are negative and edges between nodes in distinct sets are positive

- if $\quad w \in \boldsymbol{w}^k,\ k \text{ is odd} \quad \Rightarrow \quad d(w) < 0$
- if $\quad w \in \boldsymbol{w}^k,\ k \text{ is even} \quad \Rightarrow \quad d(w) > 0$



F. Harary (1957) "Structural Duality" Behavioral Sci.

Gaussian IT

Mutlivariate Gaussian system $\boldsymbol{X}$

TC: $TC(\boldsymbol{X}) = \sum_j^N \left[H(X_j)\right] - H(\boldsymbol{X}) = \frac{1}{2}\log\left[\frac{1}{\det(\Sigma)}\right]$

O-information: $\Omega(\boldsymbol{X}) = \sum_{i=1}^N TC(\boldsymbol{X}_{-i}) - (N-2)TC(\boldsymbol{X})$

$$\rightarrow \boxed{\Omega(\boldsymbol{X}) = \frac{N-2}{2}\log\left[\det(\Sigma)\right] - \sum_i^N \frac{1}{2}\log\left[\det(\Sigma_{-i})\right]}$$

Varley, T.F. et al. (2023) "Multivariate information theory uncovers synergistic subsystems of the human cerebral cortex." Commun Biol

$$\boxed{\Omega(\boldsymbol{X}) = \frac{N-2}{2}\log\left[\det(\Sigma)\right] - \sum_i^N \frac{1}{2}\log\left[\det(\Sigma_{-i})\right]}$$

Expand using: $\log\left[\det(\Sigma)\right] = -\sum_{k=2} \frac{(-1)^k}{k}\mathrm{tr}[W^k]$

where $W = \Sigma - \mathbb{I}$**,** **requires** $\rho(W) < 1$

$$\Omega(\boldsymbol{X}) = \frac{N-2}{2}\left[-\frac{\mathrm{tr}[W^2]}{2} + \frac{\mathrm{tr}[W^3]}{3} - \frac{\mathrm{tr}[W^4]}{4} + \frac{\mathrm{tr}[W^5]}{5}\dots\right]$$
$$+\frac{1}{2}\left[\frac{\sum_i \mathrm{tr}[W^2_{-i}]}{2} - \frac{\sum_i \mathrm{tr}[W^3_{-i}]}{3} + \frac{\sum_i \mathrm{tr}[W^4_{-i}]}{4} - \frac{\sum_i \mathrm{tr}[W^5_{-i}]}{5} + \dots\right]$$

- **Trick for any k:** $$\boxed{\mathrm{tr}[W^k_{-i}] = \sum_{w\in\boldsymbol{w}^k} d(w) - \sum_{w\in\boldsymbol{w}_i^k} d(w)}$$

  where $\boldsymbol{w}_i^k$ is the set of closed walks of length k that include element i at least once.

- The intuition is very similar to Lizier's "*Analytic relationship of relative synchronizability to network structure and motifs*" (2023) PNAS

L. Barnett et al (2009) "Neural complexity and structural connectivity" Physical Review E

Trick for any k: $$\boxed{\mathrm{tr}[W^k_{-i}] = \sum_{w \in \boldsymbol{w}^k} d(w) - \sum_{w \in \boldsymbol{w}_i^k} d(w)}$$

where $\boldsymbol{w}_i^k$ is the set of closed walks of length k that include element i at least once.

Then, $\sum_i \mathrm{tr}[W^k_{-i}] = N\mathrm{tr}[W^k] - \sum_{w \in \boldsymbol{w}^k} |w| d(w)$

where $|w|$ is the number of distinct elements in closed walk w.

(**Recall:** $\mathrm{tr}[W^k] = \sum_{w \in \boldsymbol{w}^k} d(w)$)

Then we can use (check again previous slide) **:**

$$\sum_i \mathrm{tr}[W^k_{-i}] = (N-2)\mathrm{tr}[W^k] - \sum_{w \in \boldsymbol{w}^k} (|w| - 2) d(w)$$

Results

$$\boxed{\Omega(\boldsymbol{X}) = \frac{1}{2}\sum_{k=3}^{\infty}\frac{1}{k}\left[(-1)^{k-1}\sum_{w\in\boldsymbol{w}^k}(|w|-2)d(w)\right]}$$

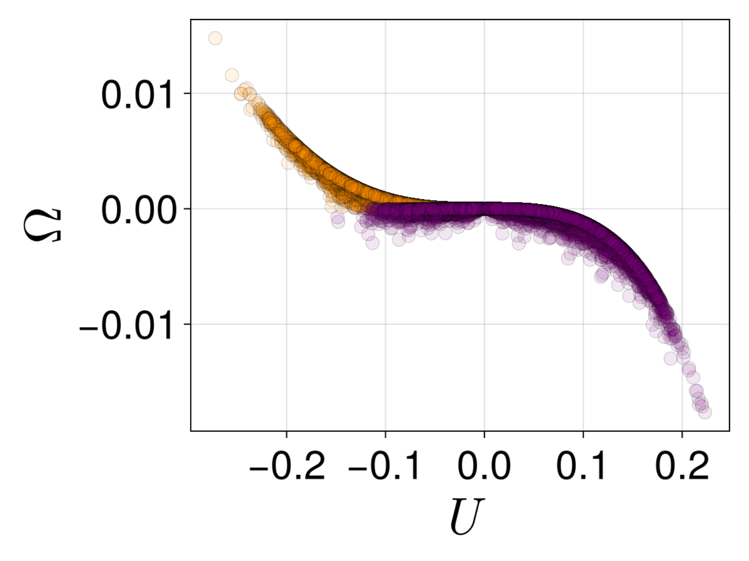
- even-length closed walks with $d(w) > 0$ and odd-length closed walks with $d(w) < 0$ **contribute synergistically**
- while even-length closed walks with $d(w) < 0$ and odd-length closed walks with $d(w) > 0$ **contribute redundantly**.

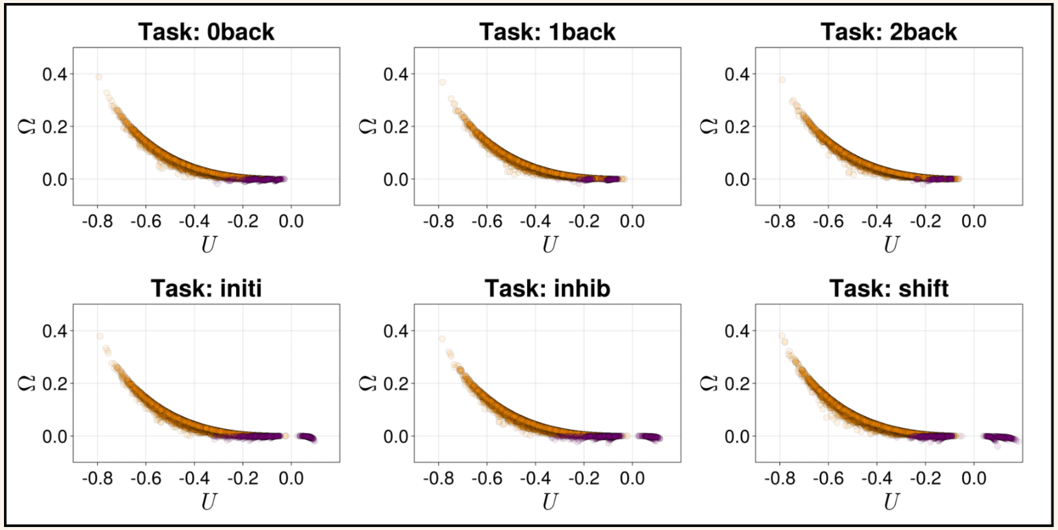
Thus, if Σ can be represented by an antibalanced graph, the system is ensured to be synergy-dominated.

From $$\boxed{\Omega(\boldsymbol{X}) = \frac{1}{2}\sum_{k=3}^{\infty}\frac{1}{k}\left[(-1)^{k-1}\sum_{w\in\boldsymbol{w}^k}(|w|-2)d(w)\right]}$$

leading terms $\sim$ structural energy metric U (from SBT)

$$U = -\binom{N}{3}^{-1}\sum_{w\in\boldsymbol{w}^3}\sqrt[3]{d(w)}$$





We now have a good idea what synergistic correlational structures look like, so

what kind of interaction matrices in linear Gaussian systems lead to these correlational structures?

Ornstein-Ulenbech Processes

$$dX(t) = -X(t) \cdot (\mathbb{I} - A)dt + d\mathcal{W}(t)$$

resulting dynamics (covariance $\Sigma^\star$):

$$\Sigma^\star = \frac{1}{2}(\mathbb{I} - A)^{-1}$$

let $A_{12} = x$, $A_{13} = y$, $A_{23} = z$ standardize $\Sigma^\star$, plug in Σ into expression for Ω and (with the help of Mathematica)

$$\Omega < 0 \quad \text{if} \quad -2xyz > (xy)^2 + (xz)^2 + (yz)^2 - (xyz)^2$$

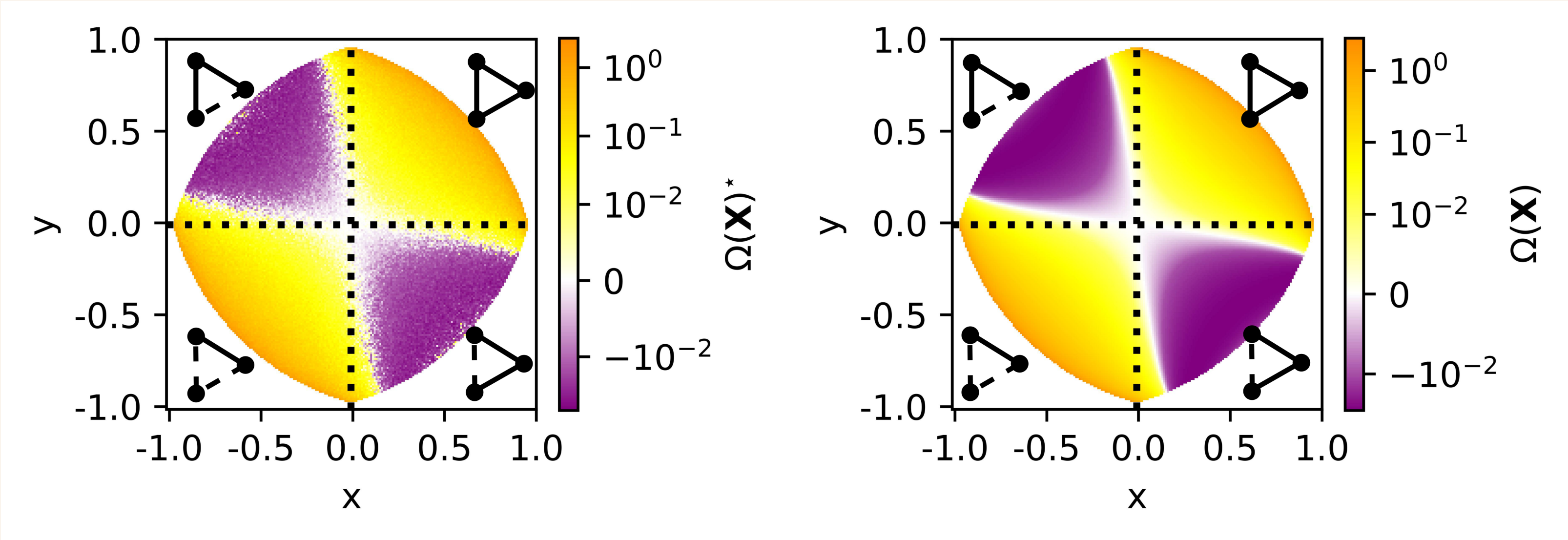
- RHS is always > 0 (if A Schur stable)
- thus we require $xyz < 0$
- A needs to be antibalanced for the system to be synergy-dominated

again, L. Barnett et al (2009) "Neural complexity and structural connectivity" Physical Review E

Sanity check

Prameter space exploration

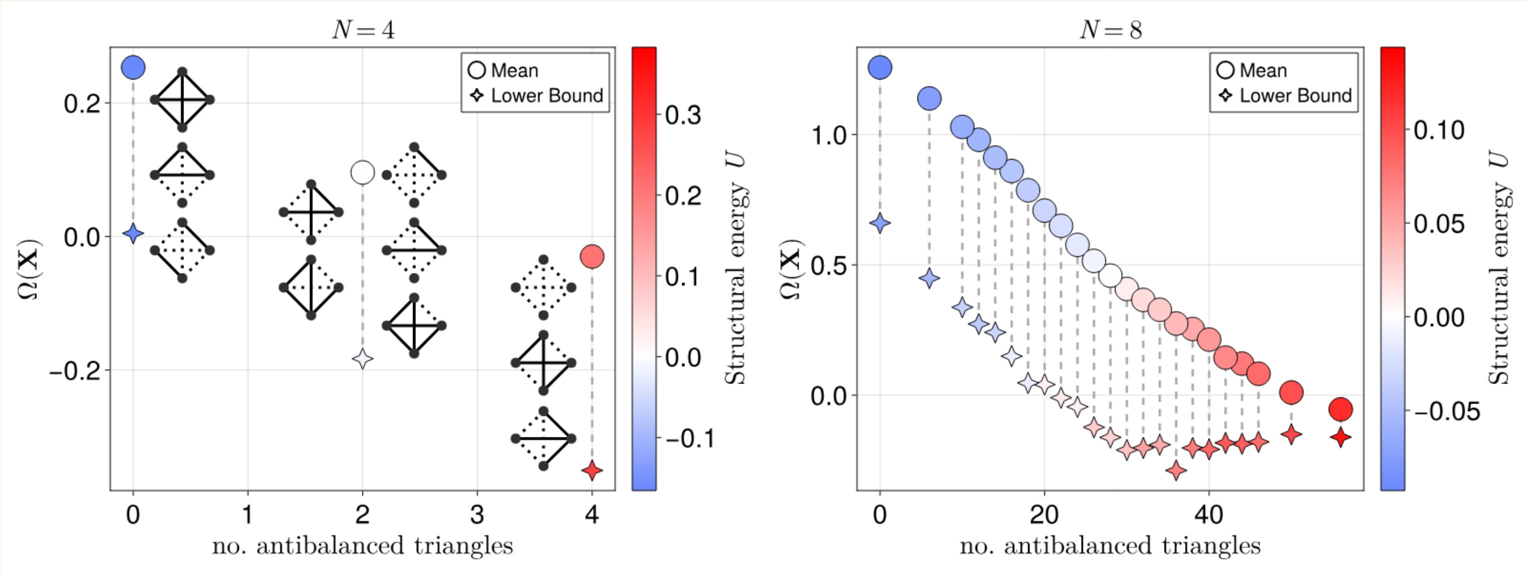
constant $z = 0.25$ (and again $A_{12} = x$, $A_{13} = y$, $A_{23} = z$)



(Left) Simulated systems Ω.

(Right) Theoretical Ω.

what about higher N? Analytically, it is a mess.



where structural energy $U = -\binom{N}{3}^{-1} \sum_{w \in \boldsymbol{w}^3} \sqrt[3]{d(w)}$

S. A. Marvel, S. H. Strogatz, and J. M. Kleinberg, (2009) Energy landscape of social balance, PRL

Conclusions

- sufficient condition for synergy-dominance in multivariate Gaussian systems of any dimension (and for redundancy)
- --> suggest SBT as an instrumental conceptual lens for studying higher-order interdependencies (see my poster! O-info and SBT in brains)
- antibalanced pairwise interactions are necessary for synergy-dominance
- --> our analytically informed interpretation is in line with observations in Ising model and oscillatory systems, e.g., D. Marinazzo et al (2019) "Synergy as a warning sign of transitions:", A. I. Luppi et al (2024) "Competitive interactions shape brain dynamics and computation across species"

Acknowledgments

Many thanks to my supervisor *Luc Berthouze*

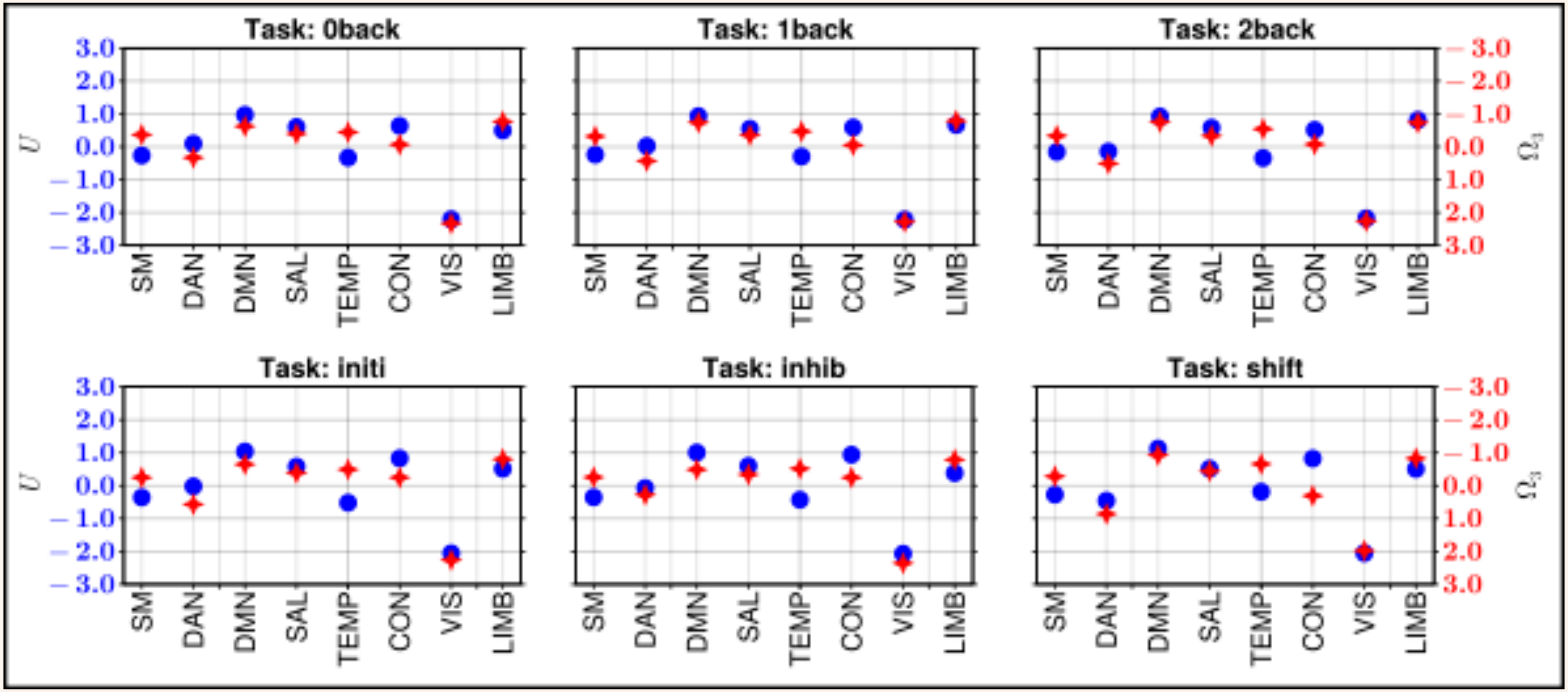
and to *Pedro Mediano*, *Adam Barrett*, and *Lionel Barnett* for the helpful discussions.

Preprint out now! **arXiv:2505.24686** **check it out and lmk**

drop me a line **ec627@sussex.ac.uk**,

check poster online here: **https://enricocaprioglio.github.io/Lucciole/publications/**

Extra



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